Medium-fi Prototyping

InForm: Embodied Haptic Feedback for Safer Lifting

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01

Value Proposition

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What is our value proposition?

inForm

A wearable multi-point haptic band that provides real-time, private vibration feedback to correct lifting form, without audio or public attention.

Designed for:

- Self-conscious gym-goers
- Casual lifters without coaching
- Users who want safer form without filming

Why this matters?

- Enables private, real-time learning of lifting form
- Reduces dependence on filming or coaching

02

Problem & Solution

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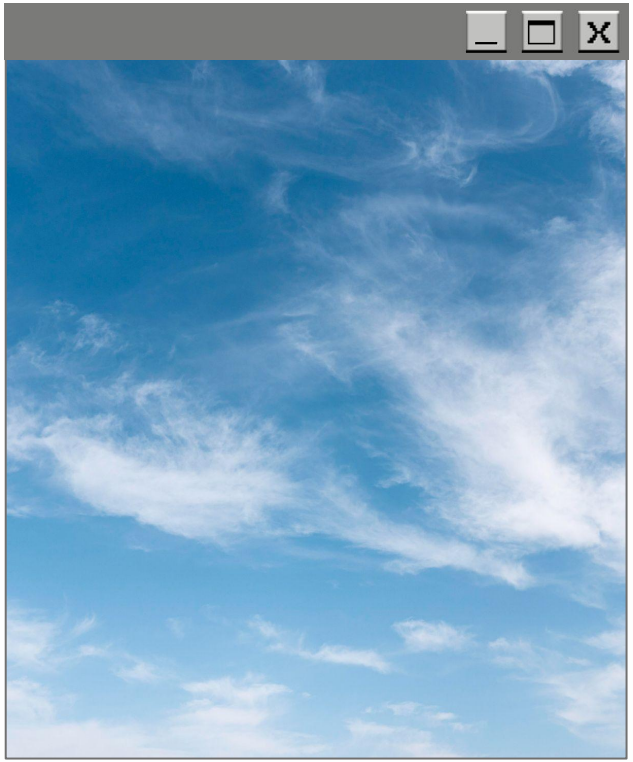

Problem

Many casual gym-goers lack real-time feedback on their form. Filming feels uncomfortable, trainers are expensive, and visible tech can increase anxiety in crowded gyms.

Key Issues:

- No real-time form correction
- Discomfort with filming
- Coaching is costly
- Anxiety in busy gyms
- Existing solutions require screens or audio
- Visible tools draw attention

Solution Preview

InForm uses multi-point haptic bands to provide private, real-time vibration feedback during lifts. Users can correct posture instantly without screens, audio, or public attention.

Core Features:

- Localized vibration feedback
- Fully private
- Preserves workout flow
- Supports learning through movement

Why haptic feedback?

- Immediate correction without visual attention
- Preserves workout flow
- Supports embodied learning

03 Tasks

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Tasks

01

Task Simple: Calibrate and correctly wear the InForm band before beginning exercise.

Why Simple?

- Linear flow
- Clear success state
- No real-time correction yet

02

Task Moderate: Respond to vibration feedback and correct posture during exercise.

Why Moderate?

- Requires interpreting haptic cue
- Requires body adjustment
- Real-time response needed

03

Task Complex: Apply learned corrections across a new exercise with reduced feedback.

Why Complex?

- Less feedback provided
- Requires memory of cues
- Tests embodied learning

The three primary tasks remain the same as GR5, but the introduction flow and feedback interactions were refined based on usability findings.

04 USABILITY GOALS

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Key Measurements

Usability Goal 1 — Low Mental Effort

Measurement:

- Self-reported effort (1–5)
- Workout interruptions during setup and exercise flow

Usability Goal 2 — Real-Time Learnability of Haptic Feedback

Measurement:

- Time to interpret feedback meaning
- Successful correction without facilitator explanation

Progress towards Goals

Low Mental Effort — GR5 finding:
Participants hesitated during tracker setup and exercise entry due to unclear system state.

GR6 change
Tracker status visibility, simplified setup flow, and clearer exercise entry were introduced.

Progress toward measurement:
Participants showed reduced hesitation and fewer workflow interruptions.

Real-Time Learnability — GR5 finding:
Participants were unsure how feedback cues related to body placement and calibration steps.

GR6 change
Calibration screens now explicitly show required joints and exercise-specific placement.

Progress toward measurement:
Participants interpreted feedback meaning faster and required fewer clarification questions.

05 Design revisions

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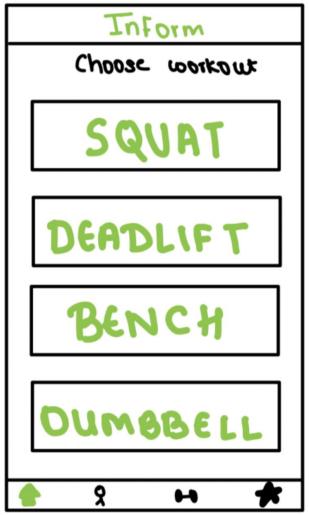
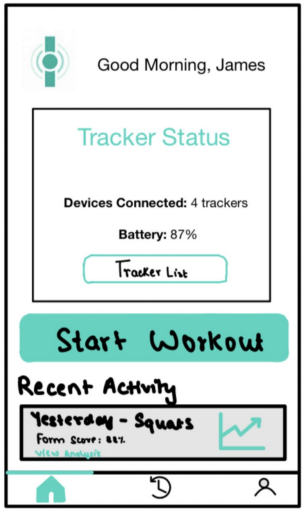
Revision 1

Previous: HomePage consisted of only a list of exercises and had no information regarding trackers.



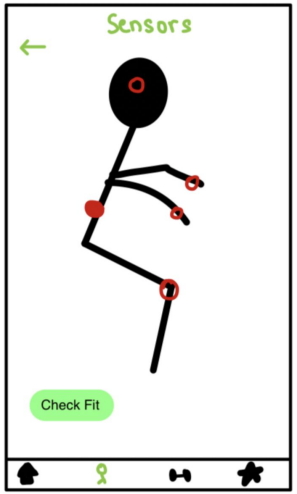
New: Our current HomePage shows tracker status with battery percentage, devices connected and history of exercises performed. Makes it easier for users to understand the app.

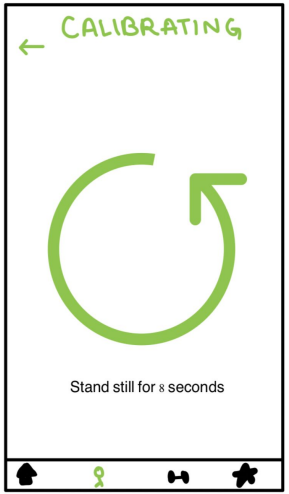
'Start Workout' button leads to a scrollable list of exercises that user wants to perform.



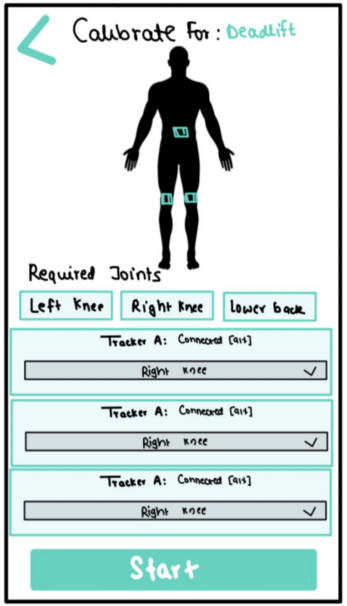
Revision 2

Previous: Joint tracker map and calibration happened in two different pages. Created confusion for the user.





New: Now, the map and calibration are on the same page. New ‘required joints’ field corresponding to the exercise you’ve chosen.

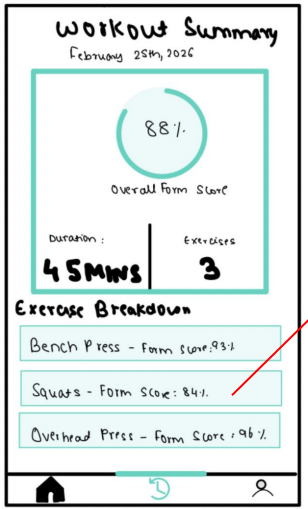


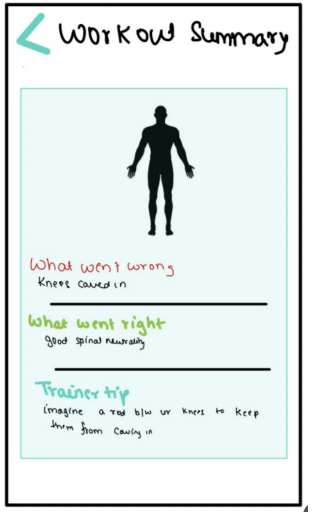
Revision 3

Previous: Post-workout summary only consisted cluttered up text. Visually unappealing, format was too difficult to navigate.

New: Workout summary now has infographics, metrics (percentage, duration, exercises) and a summary of all the exercises user performed.

The exercise breakdown buttons are clickable which leads you to workout summary.





06

Medium-Fi Task Flows

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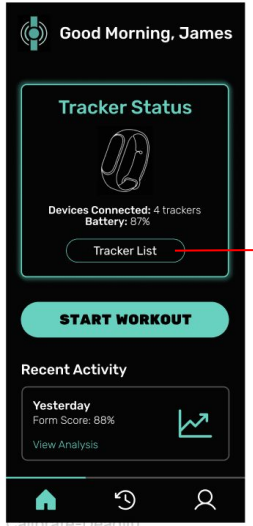
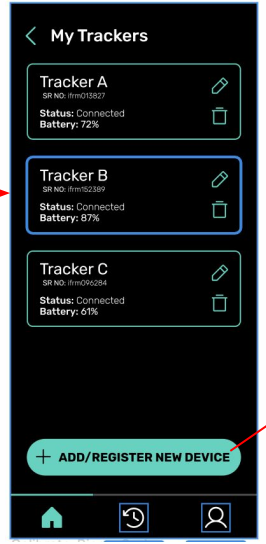
Simple Task

Recognize the incorrect lifting form in real time

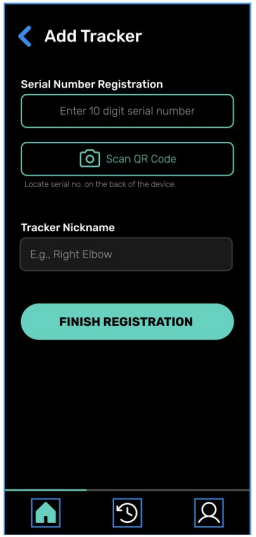
Click on “tracker list”



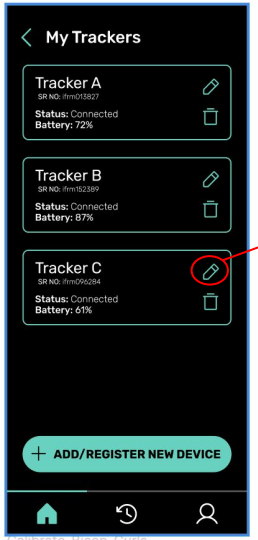
You can add a new device

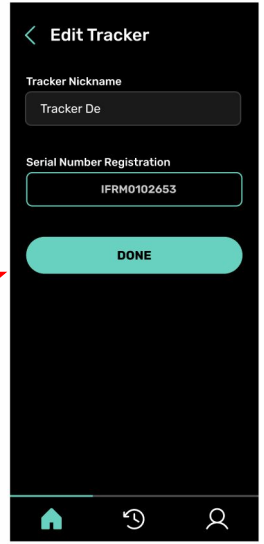


Register your new device



Edit “tracker”

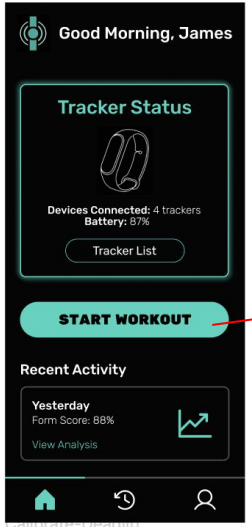


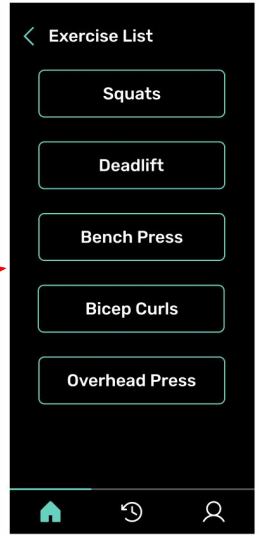


Moderate Task

Improvise the form during the particular exercise

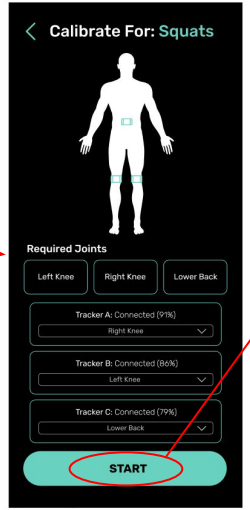
Start workout

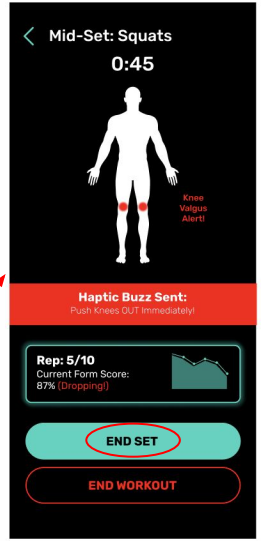




Pick an exercise (for eg, squats)

Calibrates for the particular exercise



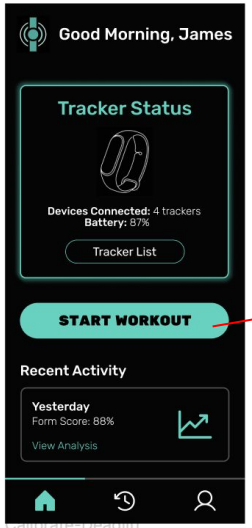


After haptics, end set.

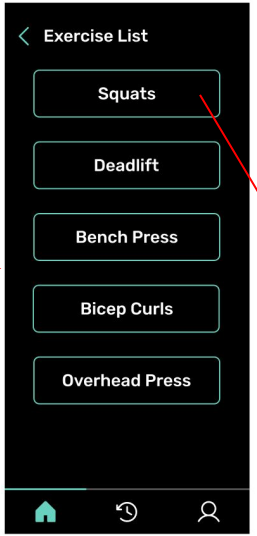
Complex Task

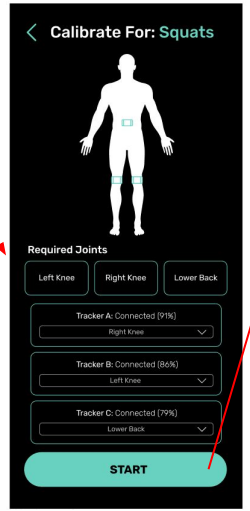
Apply correct form across different exercises

Start workout

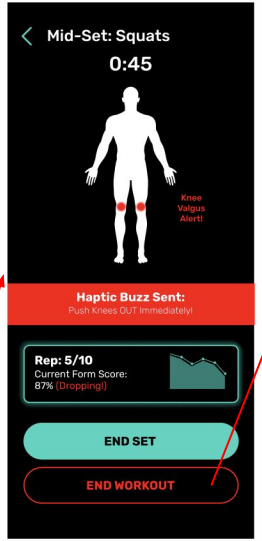


Exercise List

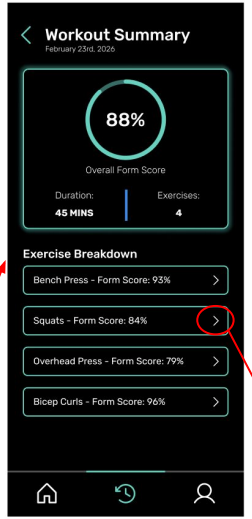




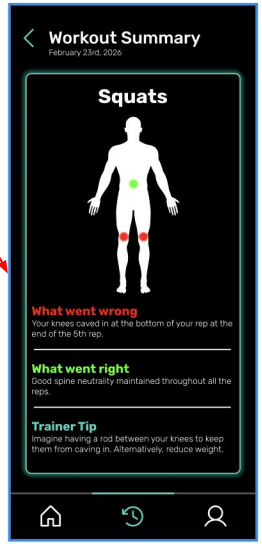
Calibration done, click start^



End workout



Get workout summary & information about improvisation



07

Prototype Implementation

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Implementation Overview

- The Medium-fi prototype was implemented in Figma using interactive flows to represent tracker setup, calibration, exercise progression, and workout summaries.
- Clickable transitions were used to simulate system responses and navigation between workout states.
- This approach allowed evaluation of usability goals without requiring functional sensing hardware.

Implementation Limitations

The prototype does not evaluate:

- A physical perception of feedback
- Real-time system latency
- Long-term learning effects

Future work should evaluate these aspects using functional hardware.

Wizard of Oz + Hard Coded Features

Wizard of Oz:

- **Simulated haptic feedback:** vibrations are triggered in the prototype to represent “form incorrect” moments (no real sensors).
- **Predefined system states:** tracker connection + battery status appear consistent to help users focus on flow.
- **Scripted session progression:** workout screens advance as if the system is detecting reps and set completion.

Hard Coded Features:

- **Calibration outcome + required joints are fixed** per selected exercise (no real pairing/placement validation).
- **Mid-set visuals are fixed:** rep counter + “form score/graph” are preset to demonstrate feedback moments.
- **Post-workout summary is fixed:** exercise breakdown + tips are example outputs (no real tracker data).
- **Tracker registration/editing is simulated** (serial/name flows saved via navigation, not real data storage)

Impact

Predefined feedback scenarios may make the interface appear easier to understand than it would be in real use, limiting our ability to fully evaluate feedback learnability.

Because tracker states and calibration outcomes were fixed, the prototype does not capture unexpected setup errors or alternative user behaviors, making it difficult to assess real mental effort during setup.

Hardcoded exercise flows reduce variability in user pathways, which may underestimate interruptions that occur in real workout situations.

Predefined workout summaries present idealized feedback explanations, limiting evaluation of how users interpret unclear or incomplete feedback.

Navigation and session progression may appear more predictable than in a live system, which can affect how intuitive the experience seems during testing.

08 Appendix

Figma Link